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CANDIDATE NUMBER

2020 Trial HSC Examination

Form VI Mathematics Extension 2

Wednesday 12th August 2020

General Instructions

- Reading time — 10 minutes
- Working time — 3 hours
- Attempt all questions.
- Write using black pen.
- Calculators approved by NESA may be used.
- A loose reference sheet is provided separate to this paper.

Total Marks: 100

Section I (10 marks) Questions 1–10

- This section is multiple-choice. Each question is worth 1 mark.
- Record your answers on the provided answer sheet.

Section II (90 marks) Questions 11–16

- Relevant mathematical reasoning and calculations are required.
- Start each question in a new booklet.

Collection

- If you use multiple booklets for a question, place them inside the first booklet for the question.
- Arrange your solutions in order.
- Write your candidate number on this page and on the multiple choice sheet.
- Place everything inside this question booklet.

Checklist

- Reference sheet
- Multiple-choice answer sheet
- 6 booklets per boy
- Candidature: 78 pupils

Writer: WJM

Section I

Questions in this section are multiple-choice.

Choose a single best answer for each question and record it on the provided answer sheet.

1. Which of the following is the converse of $\sim P \Rightarrow Q$?

- (A) $\sim Q \Rightarrow P$
- (B) $Q \Rightarrow \sim P$
- (C) $P \Rightarrow Q$
- (D) $\sim P \Rightarrow \sim Q$

2. Which of the following is a primitive of $\tan^4 2x \sec^2 2x$?

- (A) $\tan^5 2x$
- (B) $\frac{1}{2} \tan^5 2x$
- (C) $\frac{1}{5} \tan^5 2x$
- (D) $\frac{1}{10} \tan^5 2x$

3. What is the smallest positive value of θ such that $e^{i\theta} \times e^{2i\theta} = i$?

- (A) $\frac{\pi}{12}$
- (B) $\frac{\pi}{6}$
- (C) $\frac{\pi}{3}$
- (D) $\frac{5\pi}{6}$

4. What is the approximate size of the angle between the vectors $\begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$?

- (A) 57°
- (B) 93°
- (C) 123°
- (D) 158°

5. What are the zeros of the polynomial $P(x) = x^3 - 3x^2 + x + 5$?

- (A) $1, -2 + i, -2 - i$
- (B) $1, 2 + i, 2 - i$
- (C) $-1, -2 + i, -2 - i$
- (D) $-1, 2 + i, 2 - i$

6. Which expression is equivalent to $\int (\ln x)^2 dx$?

- (A) $x (\ln x)^2 - 2 \int \ln x dx$
- (B) $(\ln x)^2 - 2 \int \ln x dx$
- (C) $x (\ln x)^2 - 2 \int x \ln x dx$
- (D) $2 \int \ln x dx$

7. The displacement x of a particle in metres after t seconds is given by $x = 2 + 4 \sin^2 t$. How far will the particle travel in the first 2π seconds?

- (A) 0 metres
- (B) 2 metres
- (C) 8 metres
- (D) 16 metres

8. The polynomial $P(z)$ has real coefficients. The complex number α is of the form $a + ib$, where a and b are both real, non-zero and distinct.

If $P(a)$, $P'(a)$, $P(b)$, $P'(b)$ and $P(\alpha)$ are all zero, what is the minimum degree of $P(z)$?

- (A) 4
- (B) 5
- (C) 6
- (D) 7

9. Without evaluating the integrals, which of the following integrals has the largest value?

(A) $\int_{-\pi}^{\pi} x \cos x \, dx$

(B) $\int_{-1}^1 \ln(x^2 + 1) \, dx$

(C) $\int_0^1 (2^{-x} - 1) \, dx$

(D) $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} (\sin^{-1} x)^3 \, dx$

10. A complex number z is defined such that $|z - 1| = |z + 2 - i\sqrt{3}|$.

What is the value of $\text{Arg}(z)$ when $|z|$ is a minimum?

(A) $\frac{\pi}{6}$

(B) $\frac{\pi}{3}$

(C) $\frac{2\pi}{3}$

(D) $\frac{5\pi}{6}$

End of Section I

The paper continues in the next section

Section II

This section consists of long-answer questions.

Marks may be awarded for reasoning and calculations.

Marks may be lost for poor setting out or poor logic.

Start each question in a new booklet.

QUESTION ELEVEN (15 marks) Start a new answer booklet. Marks

(a) Express $\frac{1-8i}{2-i}$ in the form $a+ib$, where a and b are real. 2

(b) Find:

(i) $\int x \cos x \, dx$ 2

(ii) $\int \frac{dx}{x^2+4x+8}$ 2

(c) Find any values of λ for which the vectors $\begin{bmatrix} -2 \\ \lambda \\ 2\lambda \end{bmatrix}$ and $\begin{bmatrix} 4 \\ \lambda \\ -1 \end{bmatrix}$ are perpendicular. 2

(d) (i) Find the constants A , B and C such that 2

$$\frac{5x^2 - x + 5}{(x^2 + 2)(x - 1)} = \frac{Ax + B}{x^2 + 2} + \frac{C}{x - 1}.$$

(ii) Hence find $\int \frac{5x^2 - x + 5}{(x^2 + 2)(x - 1)} dx$. 2

(e) Three lines have equations: 3

$$y = px + b_1$$

$$y = qx + b_2$$

$$y = rx + b_3$$

where p, q, r, b_1, b_2 and b_3 are real constants and p, q and r are distinct.

Use proof by contradiction to show algebraically that these lines cannot be perpendicular to one another.

QUESTION TWELVE (15 marks) Start a new answer booklet.**Marks**

- (a) Sketch the region in the complex plane which simultaneously satisfies

3

$$|z| < \sqrt{2} \text{ and } 0 \leq \arg(z) \leq \frac{\pi}{4}.$$

Clearly label the coordinates of any corners of the region, indicating if they are included in the region.

- (b) Using the substitution
- $t = \tan \frac{x}{2}$
- , or otherwise, evaluate
- $\int_0^{\frac{\pi}{2}} \frac{dx}{2 + \sin x + 2 \cos x}$
- .

3

- (c) Use proof by contraposition to show that for
- $x \in \mathbf{Z}$
- , if
- $x^2 - 6x + 5$
- is even, then
- x
- is odd.

2

- (d) (i) By solving the equation
- $z^3 + 1 = 0$
- , find the three cube roots of
- -1
- .

2

- (ii) Let
- ω
- be a non-real cube root of
- -1
- . Show that
- $\omega^2 = \omega - 1$
- .

2

- (iii) Hence simplify
- $(1 - \omega)^6$
- .

1

- (e) If
- x
- and
- y
- are positive real numbers, then
- $x + y \geq 2\sqrt{xy}$
- . (Do NOT prove this.)

2

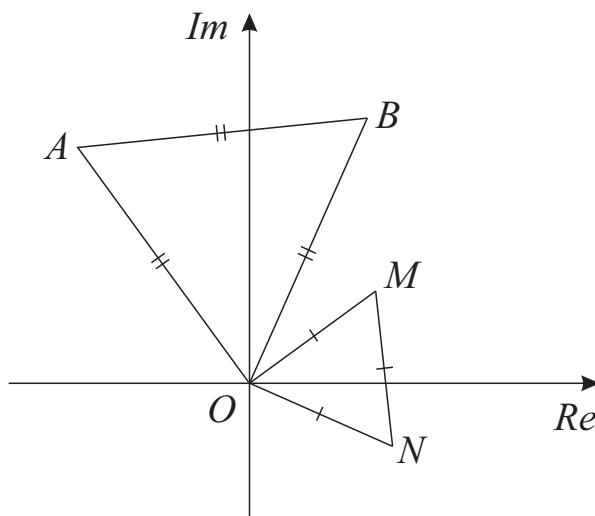
If a and b are positive real numbers, show that $(a + b) \left(\frac{1}{a} + \frac{1}{b} \right) \geq 4$.

QUESTION THIRTEEN (16 marks)

Start a new answer booklet.

Marks

(a)



The diagram above shows the points O , A , B , M and N on the complex plane. These points correspond to the complex numbers 0 , a , b , m and n respectively. The triangles OAB and OMN are equilateral. Let $\alpha = e^{\frac{i\pi}{3}}$.

(i) Explain why $m = \alpha n$.

1

(ii) Use complex numbers to show that $AM = BN$.

2

(b) Two of the zeros of $P(x) = x^4 - 12x^3 + 54x^2 - 108x + 85$ are $a + ib$ and $2a + ib$, where a and b are real and $b > 0$.

(i) Find the values of a and b .

3

(ii) Hence or otherwise express $P(x)$ as the product of quadratic factors with real coefficients.

1

(c) Two lines are defined by $\underline{v} = \begin{bmatrix} 2 \\ -1 \\ -5 \end{bmatrix} + \lambda \begin{bmatrix} 4 \\ -2 \\ -5 \end{bmatrix}$ and $\underline{u} = \begin{bmatrix} 4 \\ -3 \\ 3 \end{bmatrix} + \mu \begin{bmatrix} -5 \\ 3 \\ 1 \end{bmatrix}$, where $\lambda, \mu \in \mathbf{R}$.

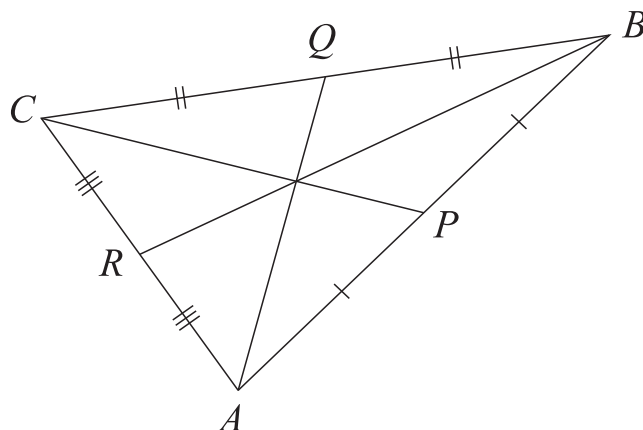
3

Show that the two lines intersect at a single point.

Question Thirteen continues over the page

QUESTION THIRTEEN (Continued)

(d)



The diagram above shows $\triangle ABC$, where A , B and C have position vectors $\underline{a}$, $\underline{b}$ and $\underline{c}$ respectively. The points P , Q and R bisect the intervals AB , BC and CA respectively.

(i) Show that $\overrightarrow{AQ} = \frac{1}{2}(\underline{c} + \underline{b}) - \underline{a}$.

1

(ii) Show that $\overrightarrow{AQ} + \overrightarrow{BR} + \overrightarrow{CP} = \underline{0}$.

2

(e) A sequence a_n is defined recursively by $a_n = a_{n-1} + 3n^2$, where $a_0 = 0$. Use mathematical induction to show that $a_n = \frac{n(n+1)(2n+1)}{2}$ for all integers $n \geq 0$.

3

QUESTION FOURTEEN (14 marks)

Start a new answer booklet.

Marks

- (a) The polynomial $P(x) = x^5 + px^4 + qx^3 + (2q - 1)x^2 + 4px + r$, where $p, q, r \in \mathbf{R}$, has a zero of $x = -1$ with multiplicity 3.

(i) Find the values of p, q and r .

3

(ii) Hence find the other zeros of $P(x)$.

2

- (b) Let $I_n = \int_0^{\frac{\pi}{2}} x^n \sin x \, dx$, for integers $n \geq 0$.

(i) Show that $I_n = n \left(\frac{\pi}{2}\right)^{n-1} - n(n-1)I_{n-2}$ for $n \geq 2$.

2

(ii) Hence evaluate $\int_0^{\frac{\pi}{2}} x^2 \sin x \, dx$.

2

- (c) Let $z = e^{i\theta}$.

(i) Show that $z^n - \frac{1}{z^n} = 2i \sin(n\theta)$.

1

(ii) Show that $\left(z - \frac{1}{z}\right)^5 = \left(z^5 - \frac{1}{z^5}\right) - 5\left(z^3 - \frac{1}{z^3}\right) + 10\left(z - \frac{1}{z}\right)$.

1

(iii) Hence find $\int \sin^5 \theta \, d\theta$.

3

QUESTION FIFTEEN (16 marks) Start a new answer booklet.**Marks**

(a) Find $\int \frac{\sqrt{x}}{1+x} dx$. **2**

(b) Use mathematical induction to show that for all **odd** integers $n \geq 1$, $4^n + 5^n + 6^n$ is divisible by 15. **3**

(c) A package with mass m kg is dropped from a stationary hovering helicopter. As the package falls vertically it experiences a force due to gravity of $10m$ Newtons. When a parachute on the package is deployed, it experiences a resistive force of magnitude mkv Newtons, where v is the velocity of the package in metres per second and k is a positive constant.

The vertical displacement of the package y metres from the position where the parachute is deployed satisfies

$$m\ddot{y} = 10m - mkv,$$

where the downwards direction is taken as positive.

(i) Let v_T be the terminal velocity of the package with the parachute deployed. Find v_T in terms of k . **1**

(ii) The parachute on the package is deployed when its velocity reaches $\frac{20}{k} \text{ ms}^{-1}$.

(α) Show that $y = \frac{1}{k^2} \left(20 - kv + 10 \ln \left| \frac{10}{10 - kv} \right| \right)$. **3**

(β) In the time that it takes the package to fall 50 m after the parachute is deployed, its velocity decreases by 25%. Find the value of k , giving your answer correct to two decimal places. **2**

(d) Two lines r_1 and r_2 have equations

$$\underline{r}_1 = \begin{bmatrix} 0 \\ 5 \\ 4 \end{bmatrix} + \lambda \begin{bmatrix} -1 \\ 4 \\ 3 \end{bmatrix} \quad \text{and} \quad \underline{r}_2 = \begin{bmatrix} -2 \\ 4 \\ 1 \end{bmatrix} + \mu \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix}, \text{ where } \lambda, \mu \in \mathbf{R}.$$

The point A lies on the first line with parameter $\lambda = p$, and the point B lies on the second line with parameter $\mu = q$.

(i) Write $\overrightarrow{AB}$ as a column vector, writing the components in terms of p and q . **1**

(ii) Calculate the value of $|\overrightarrow{AB}|$ when $\overrightarrow{AB}$ is perpendicular to both $\underline{r}_1$ and $\underline{r}_2$. **3**

(iii) State the range of values that $|\overrightarrow{AB}|$ can take as p and q vary. **1**

QUESTION SIXTEEN (14 marks) Start a new answer booklet.

Marks

- (a) (i) The function
- $f(x)$
- is continuous for all
- $x \in \mathbf{R}$
- .

2

Use the substitution $x = \pi - u$ to show that

$$\int_0^\pi x f(\sin x) dx = \frac{\pi}{2} \int_0^\pi f(\sin x) dx.$$

- (ii) Hence evaluate
- $\int_0^\pi (1 + 2x) \frac{\sin^3 x}{1 + \cos^2 x} dx$
- .

3

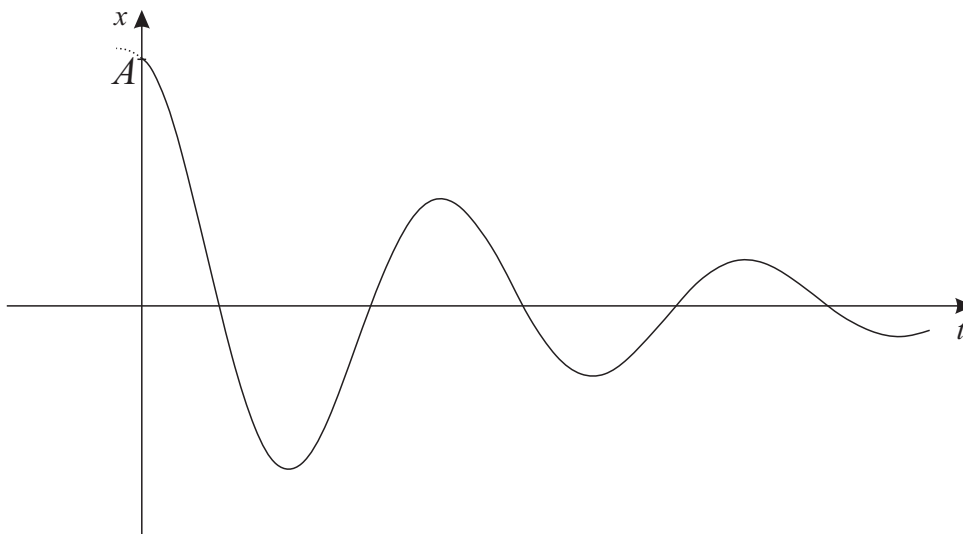
- (b) An object of unit mass is attached to a spring. When the object is pulled and released, it experiences a force proportional to its displacement
- x
- metres, where
- $x = 0$
- is taken as the centre of motion. The object moves in simple harmonic motion and the acceleration of the object is given by
- $\ddot{x} = -P^2 x$
- for some constant
- $P > 0$
- .

When the spring and object are submerged in a liquid, the object also experiences a resistive force proportional to its velocity. Thus, the acceleration of the object is given by

$$\ddot{x} = -P^2 x - Q\dot{x} \quad (*)$$

for some constant $Q > 0$.

The spring is stretched and the object is released. A timer is started once the object reaches $x = A$, where $A > 0$. That is, $x = A$ when $t = 0$. A graph of the displacement of the object submerged in liquid after t seconds is shown as follows:



The following questions relate to the motion of the object while it is submerged in liquid and $t \geq 0$.

- (i) Show that
- $x = Ae^{-kt} \cos nt$
- is a solution to the differential equation (*) if
- $k = \frac{1}{2}Q$
- and
- $n = \frac{1}{2}\sqrt{4P^2 - Q^2}$
- . You may assume that
- $4P^2 - Q^2 > 0$
- .

3

- (ii) Let
- x_r
- be the displacement of the object the
- r
- th time that it is instantaneously at rest.

2

Show that $x_1 = -Ae^{\frac{k\alpha}{n}} \cos \alpha \times e^{-\frac{k\pi}{n}}$, where $\alpha = \tan^{-1} \left(\frac{k}{n} \right)$.

- (iii) The value of the coefficient
- P
- relates to the stiffness of the spring, while the value of the coefficient
- Q
- relates to the viscosity of the liquid.

4

Show that the total distance that the object will move while submerged in a liquid for $t \geq 0$ is dependent only on the value of the ratio $\frac{P}{Q}$.

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SYDNEY GRAMMAR SCHOOL



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Wednesday 12th August 2020

- Fill in the circle completely.
- Each question has only one correct answer.

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

Question Eight

A ☐ B ☐ C ☐ D ☐

Question Nine

A ☐ B ☐ C ☐ D ☐

Question Ten

A ☐ B ☐ C ☐ D ☐

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Maths Ext.2 Trial - Solutions

MC

① $Q \Rightarrow \sim P$ is the converse of $\sim P \Rightarrow Q$
 $\therefore$ (B) ✓

②
$$\int \tan^4 2x \sec^2 2x dx = \frac{1}{2} \int \tan^4 2x \times \frac{d}{dx}(\tan 2x) dx$$
$$= \frac{1}{10} \tan^5 2x + c$$
$$\therefore$$
 (D) ✓

③
$$e^{i\theta} \times e^{2i\theta} = e^{3i\theta} \quad i = e^{i\frac{\pi}{2}}$$
$$\Rightarrow 3i\theta = i\frac{\pi}{2}$$
$$\theta = \frac{\pi}{6}$$
$$\therefore$$
 (B) ✓

④ Let $\underline{a} = \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix}$, $\underline{b} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$.

$$\underline{a} \cdot \underline{b} = 2 - 6 - 1 = -5$$
$$|\underline{a}| = \sqrt{4 + 9 + 1} = \sqrt{14}$$
$$|\underline{b}| = \sqrt{1 + 4 + 1} = \sqrt{6}$$
$$\cos \theta = \frac{\underline{a} \cdot \underline{b}}{|\underline{a}| \times |\underline{b}|} = \frac{-5}{\sqrt{14} \times \sqrt{6}}$$
$$\theta \doteq 123^\circ$$

$\therefore$ (C) ✓

⑤ Sum of zeros must be 3, ruling out A & C
Product of zeros must be -5
 $-1 \times (2+i)(2-i) = 4 - i^2 = -5$

$\therefore$ (D) ✓

$$\begin{aligned}
 \textcircled{6} \quad \int (\ln x)^2 dx &= \int (\ln x)^2 \times \frac{d}{dx}(x) dx \\
 &= x(\ln x)^2 - \int 2 \ln x \times \frac{1}{x} \times x dx \\
 &= x(\ln x)^2 - 2 \int \ln x dx \\
 \therefore \textcircled{A} \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{7} \quad x &= 2 + 4\sin^2 t \\
 &= 2 + 4 \times \frac{1}{2}(1 - \cos 2t) \\
 &= 4 - 2\cos 2t
 \end{aligned}$$

$$\text{Period} = \frac{2\pi}{2}$$

$$= \pi$$

$\therefore$ In 2π seconds, particle travels 2 full cycles with amplitude 2 metres.



$$2 \times 8 = 16$$

$$\therefore \textcircled{D} \quad \checkmark$$

$\textcircled{8} \quad P(a) = P'(a) = 0$, so a is a double zero
 $P(b) = P'(b) = 0$, so b is a double zero.
 Since $P(z)$ has real coefficients and $P(\alpha) = 0$, $P(\bar{\alpha}) = 0$.
 $\therefore \alpha$ and $\bar{\alpha}$ are zeros

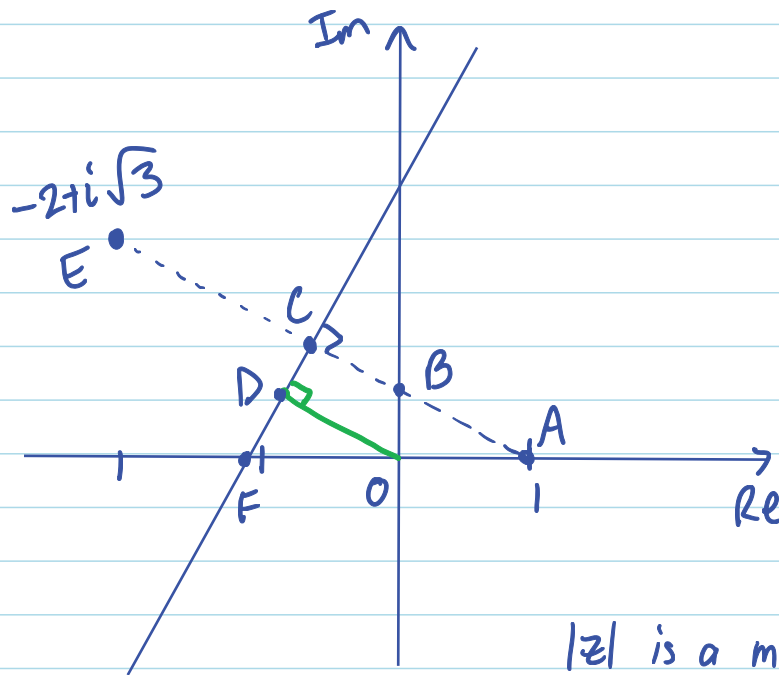
$\therefore$ minimum degree is 6

$$\therefore \textcircled{C} \quad \checkmark$$

$$\begin{aligned}
 \textcircled{9} \quad x \cos x &\rightarrow \text{odd} \\
 (\sin^{-1} x)^3 &\rightarrow \text{odd} \\
 2^{-x} - 1 &\leq 0 \quad \text{for } 0 \leq x \leq 4 \\
 \ln(x^2 + 1) &\geq 0 \quad \text{for } -1 \leq x \leq 1 \\
 \therefore \textcircled{B} \quad \checkmark
 \end{aligned}$$

(10)

z lies on the perpendicular bisector between 1 and $-2+i\sqrt{3}$



$$m_{AE} = -\frac{\sqrt{3}}{3}$$

$$= -\frac{1}{\sqrt{3}}$$

$$\therefore m_{CD} = \sqrt{3}$$

$$\Rightarrow \angle CDO = \frac{\pi}{3} \text{ and}$$

$$\angle OAB = \frac{\pi}{6}$$

$|z|$ is a minimum when z can be represented by D , where $OD \perp FC$.
 $\therefore OD \parallel AB$ and $\arg(z) = \frac{5\pi}{6}$ when $|z|$ is a minimum.

$\therefore$ (D) ✓

$$(1) (a) \quad \frac{1-8i}{2-i} \times \frac{2+i}{2+i} = \frac{2+i-16i+8}{4+1}$$

$$= 2-3i$$

$$(b) (i) \quad \int x \cos x \, dx = \int x \cdot \frac{d}{dx} (\sin x) \, dx$$

$$= x \sin x - \int \sin x \, dx$$

$$= x \sin x + \cos x + c$$

$$(ii) \quad \int \frac{dx}{x^2+4x+8} = \int \frac{dx}{(x+2)^2+4}$$

$$= \frac{1}{2} \tan^{-1} \left(\frac{x+2}{2} \right) + c$$

$$(c) \quad \begin{bmatrix} -2 \\ \lambda \\ 2\lambda \end{bmatrix} \cdot \begin{bmatrix} 4 \\ \lambda \\ -1 \end{bmatrix} = 0$$

$$\therefore -8 + \lambda^2 - 2\lambda = 0$$

$$\lambda^2 - 2\lambda - 8 = 0$$

$$(\lambda - 4)(\lambda + 2) = 0$$

$$\therefore \lambda = 4 \text{ or } -2$$

$$(d) (i) \quad \frac{5x^2 - x + 5}{(x^2+2)(x-1)} = \frac{Ax+B}{x^2+2} + \frac{C}{x-1}$$

$$5x^2 - x + 5 = (Ax+B)(x-1) + C(x^2+2)$$

$$\text{Let } x=1: 5-1+5 = 3C$$

$$C = 3$$

$$\text{Equate coef. of } x^2: 5 = A+C$$

$$\therefore A=2$$

$$\text{Let } x=0:$$

$$5 = -B + 2C$$

$$B = 1$$

✓ (any 1 value)

✓ (all values)

$$(ii) \quad \int \frac{5x^2 - x + 5}{(x^2+2)(x-1)} \, dx = \int \left(\frac{2x+1}{x^2+2} + \frac{3}{x-1} \right) \, dx$$

$$= \int \left(\frac{2x}{x^2+2} + \frac{1}{x^2+2} + \frac{3}{x-1} \right) \, dx$$

$$= \ln(x^2+2) + \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{x}{\sqrt{2}} \right) + 3 \ln|x-1| + c$$

✓ (any one correct integral)

⑪ (e) Assume that the lines $y = px + b_1$, $y = qx + b_2$, $y = rx + b_3$ are perpendicular.

Then

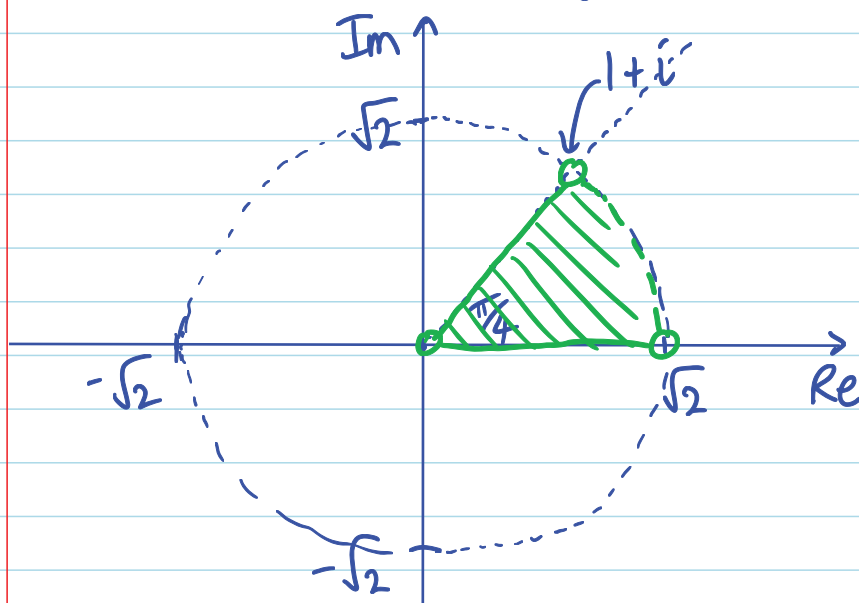
$$\begin{array}{ll} pq = -1 & \textcircled{1} \\ pr = -1 & \textcircled{2} \\ qr = -1 & \textcircled{3} \end{array} \quad \checkmark$$

① $\times$ ②: $p^2qr = 1$ $\textcircled{4}$ $\checkmark$

Sub. ③ into ④: $p^2x - 1 = 1$
 $p^2 = -1$
 $\Rightarrow$ contradiction, since $p \in \mathbb{R}$ $\checkmark$

$\therefore$ 3 lines of the form $y = mx + b$ can't be perpendicular.

(12) (a) $|z| < \sqrt{2}$, $0 \leq \arg(z) \leq \frac{\pi}{4}$



✓ one mark for correctly identifying each region

✓ correct intersection, including boundaries.

(b)

$$\int_0^{\frac{\pi}{2}} \frac{dx}{2 + \sin x + 2\cos x}$$

Let $t = \tan \frac{x}{2}$

$$\frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2}$$

$$= \frac{1}{2} (1 + \tan^2 \frac{x}{2})$$

$$= \int_0^1 \frac{2dt}{2 + \frac{2t}{1+t^2} + \frac{2(1-t^2)}{1+t^2}}$$

$$\therefore dx = \frac{2dt}{1+t^2}$$

When $x=0$, $t=0$
 $x=\frac{\pi}{2}$, $t=1$

$$= \int_0^1 \frac{2dt}{2+2t^2+2t+2-2t^2}$$

$$= \int_0^1 \frac{2dt}{2t+4}$$

$$= \int_0^1 \frac{dt}{t+2}$$

$$= [\ln(t+2)]_0^1$$

$$= \ln 3 - \ln 2$$

(c) Proof by contraposition: If x is even, then $x^2 - 6x + 5$ is odd.
Let $x = 2n$, where $n \in \mathbb{Z}^+$ so that x is even.

$$\begin{aligned}x^2 - 6x + 5 &= (2n)^2 - 6(2n) + 5 \\&= 4n^2 - 12n + 5 \\&= 2(2n^2 - 6n + 2) + 1 \\&= 2M + 1, \text{ where } M \text{ is an integer.} \\&\text{which is odd.}\end{aligned}$$

correctly identifying
contrapositive

$\therefore$ by contraposition, if $x^2 - 6x + 5$ is even, x is odd.

(d)(i) $z^3 + 1 = 0$

Let $z = \text{cis } \theta$

then by De Moivre's Theorem:

$$\text{cis } 3\theta = -1$$

$$= \text{cis } \pi$$

$$\therefore 3\theta = \pi, 3\pi, 5\pi$$

$$\theta = \frac{\pi}{3}, \pi, \frac{5\pi}{3}$$

So the cube roots of -1 are $\text{cis } \frac{\pi}{3}$, $\text{cis } \pi = -1$, $\text{cis } \frac{5\pi}{3}$

(ii) $z^3 + 1 = 0$

$$(z+1)(z^2 - z + 1) = 0$$

As -1 is a zero of $z+1$, if $w = \text{cis } \frac{\pi}{3}$ or $\text{cis } \frac{5\pi}{3}$,

$$w^2 - w + 1 = 0$$

$$w^2 = w - 1$$

(iii) $(1-w)^6 = (- (w-1))^6$

$$= (-w^2)^6$$

$$= w^{12}$$

$$= (w^3)^4$$

$$= (-1)^4$$

$$= 1$$

(e) $x + y \geq 2\sqrt{xy}$

Let $x=a, y=b$: $a+b \geq 2\sqrt{ab}$ ① ✓

Let $x=\frac{1}{a}, y=\frac{1}{b}$: $\frac{1}{a} + \frac{1}{b} \geq 2\sqrt{\frac{1}{a} \times \frac{1}{b}}$

$\frac{1}{a} + \frac{1}{b} \geq \frac{2}{\sqrt{ab}}$ ②

① $\times$ ②: $(a+b)(\frac{1}{a} + \frac{1}{b}) \geq 2\sqrt{ab} \times \frac{2}{\sqrt{ab}}$

≥ 4 ✓

(13)(a)(i) As $\triangle OMN$ is equilateral, $\angle MON = \frac{\pi}{3}$ and $|m| = |n|$.

So m can be obtained by rotating n anti-clockwise about the origin by $\frac{\pi}{3}$.
i.e. $m = n \times \text{cis } \frac{\pi}{3}$
 $= \alpha n$

(ii) $m = \alpha n$
Similarly, $a = \alpha b$.

must be specific
about nature of
rotation

$$\begin{aligned} AM &= |a - m| \\ &= |\alpha b - \alpha n| \\ &= |\alpha| \times |b - n| \\ &= 1 \times BN \end{aligned}$$

$$\therefore AM = BN$$

(b) $P(x) = x^4 - 12x^3 + 54x^2 - 108x + 85$

(i) As the coefficients of $P(z)$ are real, $a - ib$ and $2a - ib$ are also zeros of $P(z)$.

$$\text{Sum of zeros: } (a+ib) + (a-ib) + (2a+ib) + (2a-ib) = -\frac{-12}{1}$$
$$6a = 12$$

$$a = 2$$

$$\text{Product of zeros: } (a+ib)(a-ib)(2a+ib)(2a-ib) = \frac{85}{1}$$

$$(a^2 + b^2)(4a^2 + b^2) = 85$$

$$4a^4 + 5a^2b^2 + b^4 = 85$$

$$b^4 + 20b^2 + 64 = 85$$

$$b^4 + 20b^2 - 21 = 0$$

$$(b^2 + 21)(b^2 - 1) = 0$$

Since b is real and $b > 0$, $b = 1$.

$$\therefore a = 2, b = 1$$

(ii) $P(x) = (x - (2+ib))(x - (2-ib))(x - (4+ib))(x - (4-ib))$
 $= (x^2 - 4x + 5)(x^2 - 8x + 17)$

⑬(c)

$$\underline{v} = \begin{bmatrix} 2+4\lambda \\ -1-2\lambda \\ -5-5\lambda \end{bmatrix}, \quad \underline{u} = \begin{bmatrix} 4-5\mu \\ -3+3\mu \\ 3+\mu \end{bmatrix}$$

Lines intersect if a solution to the system of equations exist:

$$\begin{aligned} 2+4\lambda &= 4-5\mu \\ 4\lambda+5\mu-2 &= 0 \end{aligned} \quad \textcircled{1}$$

$$\begin{aligned} -1-2\lambda &= -3+3\mu \\ -2\lambda-3\mu+2 &= 0 \end{aligned} \quad \textcircled{2}$$

$$\begin{aligned} -5-5\lambda &= 3+\mu \\ 5\lambda+\mu+8 &= 0 \end{aligned} \quad \textcircled{3}$$

$$\textcircled{1} + 2 \times \textcircled{2}: \quad \begin{aligned} -\mu + 2 &= 0 \\ \mu &= 2 \end{aligned}$$

$$\text{sub. into } \textcircled{1}: \quad \begin{aligned} 4\lambda + 10 - 2 &= 0 \\ \lambda &= -2 \end{aligned}$$

$$\text{Check } \textcircled{3}: \quad \begin{aligned} \text{LHS} &= 5 \times -2 + 2 + 8 \\ &= 0 \end{aligned}$$

As the solution to $\textcircled{1}$ and $\textcircled{2}$ satisfies $\textcircled{3}$, the lines intersect.

⑬ (d)(i)

$$\begin{aligned}\vec{AQ} &= \vec{AB} + \vec{BQ} \\ &= (b-a) + \frac{1}{2}(c-b) \\ &= b-a + \frac{1}{2}c - \frac{1}{2}b \\ &= \frac{1}{2}(b+c) - a\end{aligned}$$

(ii)

$$\begin{aligned}\text{Similarly, } \vec{BR} &= \frac{1}{2}(a+c) - b \\ \vec{CP} &= \frac{1}{2}(a+b) - c\end{aligned}$$

$$\begin{aligned}\vec{AQ} + \vec{BR} + \vec{CP} &= \frac{1}{2}(b+c) - a + \frac{1}{2}(a+c) - b + \frac{1}{2}(a+b) - c \\ &= \underline{0}\end{aligned}$$

(e)

$$a_n = a_{n-1} + 3n^2, \quad a_0 = 0. \quad a_n = \frac{n(n+1)(2n+1)}{2}$$

Prove true for $n=0$: $a_0 = 0$ (given)

$$\begin{aligned}\text{Using formula: } a_0 &= \frac{0(0+1)(2 \times 0 + 1)}{2} \\ &= 0\end{aligned}$$

$\therefore$ true for $n=0$

$$\text{Assume true for some } n=k, \text{ i.e. } a_k = \frac{k(k+1)(2k+1)}{2}$$

Prove true for $n=k+1$:

$$\begin{aligned}\text{RTP: } a_{k+1} &= \frac{(k+1)(k+2)(2(k+1)+1)}{2} \\ &= \frac{(k+1)(k+2)(2k+3)}{2}\end{aligned}$$

$$\begin{aligned}\text{LHS} &= a_{k+1} \\ &= a_k + 3(k+1)^2 \quad (\text{from definition}) \\ &= \frac{k(k+1)(2k+1)}{2} + 3(k+1)^2 \quad (\text{from assumption})\end{aligned}$$

⑬(e) contd.

$$= \frac{k(k+1)(2k+1) + 6(k+1)^2}{2}$$

$$= \frac{(k+1)[k(2k+1) + 6(k+1)]}{2}$$

$$= \frac{(k+1)(2k^2 + 7k + 6)}{2}$$

$$= \frac{(k+1)(k+2)(2k+3)}{2}$$



$$= \text{RHS}$$

$\therefore a_n = \frac{n(n+1)(2n+1)}{2}$ for integers $n \geq 1$ by mathematical induction.

14(a)(i)

$$P(x) = x^5 + px^4 + qx^3 + (2q-1)x^2 + 4px + r$$

$$P'(x) = 5x^4 + 4px^3 + 3qx^2 + 2(2q-1)x + 4p$$

$$P''(x) = 20x^3 + 12px^2 + 6qx + 2(2q-1)$$

$$P''(-1) = 0: \quad -20 + 12p - 6q + 4q - 2 = 0$$

$$12p - 2q = 22$$

$$6p - q = 11$$

$$P'(-1) = 0:$$

$$5 - 4p + 3q - 2(2q-1) + 4p = 0$$

$$5 + 3q - 4q + 2 = 0$$

$$q = 7$$

sub. into ①:

$$6p - 7 = 11$$

$$6p = 18$$

$$p = 3$$

$$P(-1) = 0: \quad -1 + p - q + (2q-1) - 4p + r = 0$$

$$-1 + 3 - 7 + 2 \times 7 - 1 - 4 \times 3 + r = 0$$

$$r = 4$$

(ii) The coefficients of $P(x)$ are real, so let the zeros be $-1, -1, -1, a+ib, a-ib, a, b \in \mathbb{R}$.

$$\text{Sum of zeros: } -3 + 2a = -3$$

$$a = 0$$

$$\text{Product of zeros: } (-1)^3 \times (a+ib)(a-ib) = -4$$

$$-(a^2 + b^2) = -4$$

$$b^2 = 4 \quad (a=0)$$

$$b = \pm 2$$

$\therefore$ the other zeros are $2i$ and $-2i$

14(a)

Alternate solution

$$P(x) = x^5 + px^4 + qx^3 + (2q-1)x^2 + 4px + r$$

Let the zeros be $\alpha, \beta, -1, -1, -1$.

Product of zeros:

$$\begin{aligned} -\alpha\beta &= -r \\ \alpha\beta &= r \end{aligned} \quad (1)$$

Zeros four at a time:

$$\begin{aligned} \alpha\beta + \alpha\beta + \alpha\beta - \alpha - \beta &= 4p \\ 3\alpha\beta - \alpha - \beta &= 4p \end{aligned} \quad (2)$$

Zeros three at a time:

$$\begin{aligned} -\alpha\beta - \alpha\beta - \alpha\beta + \alpha + \alpha + \alpha + \beta + \beta + \beta - 1 &= -(2q-1) \\ -3\alpha\beta + 3\alpha + 3\beta - 1 &= -(2q-1) \\ 3\alpha\beta - 3\alpha - 3\beta + 1 &= 2q-1 \end{aligned} \quad (3)$$

Zeros two at a time:

$$\begin{aligned} \alpha\beta - \alpha - \alpha - \alpha - \beta - \beta - \beta + 1 + 1 + 1 &= q \\ \alpha\beta - 3\alpha - 3\beta + 3 &= q \end{aligned} \quad (4)$$

Sum of zeros:

$$\alpha + \beta - 3 = -p \quad (5)$$

(2) + 4 × (5): $3\alpha\beta - \alpha - \beta + 4(\alpha + \beta - 3) = 0$

$$\alpha\beta + \alpha + \beta - 4 = 0 \quad (6)$$

sub. (4) into (3):

$$3\alpha\beta - 3\alpha - 3\beta + 1 = 2(\alpha\beta - 3\alpha - 3\beta + 3) - 1$$

$$\alpha\beta + 3\alpha + 3\beta - 4 = 0 \quad (7)$$

(7) - (6):

$$2\alpha + 2\beta = 0$$

$$\therefore \alpha + \beta = 0 \quad (8)$$

sub. into (6):

$$\alpha\beta - 4 = 0$$

$$\alpha\beta = 4 \quad (9)$$

Substituting (8) and (9) into (1), (4), and (5) gives:

$$r = 4$$

$$q = 4 - 3 \times 0 + 3 = 7$$

$$p = 3 - 0 = 3$$

(ii) From (8): $\beta = -\alpha$

sub. into (9): $-\alpha^2 = 4$ ✓

$$\alpha^2 = -4$$

$$\alpha = \pm 2i$$

$\therefore$ the other zeros are $2i$ and $-2i$ ✓

Correctly obtaining
4 equations ✓

(14) (b)(i)

$$\begin{aligned} I_n &= \int_0^{\frac{\pi}{2}} x^n \sin x \, dx \\ &= \int_0^{\frac{\pi}{2}} x^n \cdot \frac{d}{dx}(-\cos x) \, dx \\ &= \left[-x^n \cos x \right]_0^{\frac{\pi}{2}} + n \int_0^{\frac{\pi}{2}} x^{n-1} \cos x \, dx \quad \checkmark \\ &= 0 + n \int_0^{\frac{\pi}{2}} x^{n-1} \times \frac{d}{dx}(\sin x) \, dx \\ &= n \left(\left[x^{n-1} \sin x \right]_0^{\frac{\pi}{2}} - (n-1) \int_0^{\frac{\pi}{2}} x^{n-2} \sin x \, dx \right) \end{aligned}$$

$$\begin{aligned} \therefore I_n &= n \left(\left(\frac{\pi}{2} \right)^{n-1} - (n-1) I_{n-2} \right) \\ &= n \left(\frac{\pi}{2} \right)^{n-1} - n(n-1) I_{n-2} \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad I_0 &= \int_0^{\frac{\pi}{2}} \sin x \, dx \\ &= \left[-\cos x \right]_0^{\frac{\pi}{2}} \\ &= 0 - (-1) \\ &= 1 \quad \checkmark \end{aligned}$$

$$\begin{aligned} \int_0^{\frac{\pi}{2}} x^2 \sin x \, dx &= I_2 \\ &= 2 \left(\frac{\pi}{2} \right)^{2-1} - 2 \times 1 \times 1 \quad \checkmark \\ &= \pi - 2 \end{aligned}$$

$$\begin{aligned} \text{(c)(i)} \quad z^n - \frac{1}{z^n} &= e^{in\theta} - \frac{1}{e^{in\theta}} \quad (\text{by De Moivre's theorem}) \\ &= e^{in\theta} - e^{-in\theta} \end{aligned}$$

$$\begin{aligned} &= \cos(n\theta) + i\sin(n\theta) - (\cos(-n\theta) + i\sin(-n\theta)) \\ &= \cos(n\theta) + i\sin(n\theta) - \cos(n\theta) + i\sin(n\theta) \\ &= 2i\sin(n\theta) \quad \checkmark \text{ as cosine even, sine odd.} \end{aligned}$$

(relatively lenient here)

(14) (c) (ii)

$$\begin{aligned}
 \left(z - \frac{1}{z}\right)^5 &= \binom{5}{0} z^5 + \binom{5}{1} z^4 \left(-\frac{1}{z}\right)^1 + \binom{5}{2} z^3 \left(-\frac{1}{z}\right)^2 + \binom{5}{3} z^2 \left(-\frac{1}{z}\right)^3 \\
 &\quad + \binom{5}{4} z \left(-\frac{1}{z}\right)^4 + \binom{5}{5} \left(-\frac{1}{z}\right)^5 \\
 &= z^5 - 5z^3 + 10z - 10 \times \frac{1}{z} + 5 \times \frac{1}{z^3} - \frac{1}{z^5} \\
 &= \left(z^5 - \frac{1}{z^5}\right) - 5\left(z^3 - \frac{1}{z^3}\right) + 10\left(z - \frac{1}{z}\right) \quad \checkmark
 \end{aligned}$$

(iii) Substituting result from part (i) into identity in part (ii):

$$(2i \sin \theta)^5 = 2i \sin 5\theta - 5(2i \sin 3\theta) + 10(2i \sin \theta) \quad \checkmark$$

$$32i \sin^5 \theta = 2i \sin 5\theta - 10i \sin 3\theta + 20i \sin \theta$$

$$\therefore \sin^5 \theta = \frac{1}{16} (\sin 5\theta - 5 \sin 3\theta + 10 \sin \theta) \quad \checkmark$$

So $\int \sin^5 \theta d\theta = \frac{1}{16} \left(-\frac{1}{5} \cos 5\theta + \frac{5}{3} \cos 3\theta - 10 \cos \theta \right) + C \quad \checkmark$

$$= -\frac{1}{80} \cos 5\theta + \frac{5}{48} \cos 3\theta - \frac{5}{8} \cos \theta + C$$

15 (a)

$$\int \frac{\sqrt{x} dx}{1+x}$$

$$\begin{aligned} \text{Let } u &= \sqrt{x} \\ u^2 &= x \\ 2u du &= dx \end{aligned}$$

$$= \int \frac{u \times 2u du}{1+u^2} \quad \checkmark$$

$$= 2 \int \frac{u^2 du}{1+u^2}$$

$$= 2 \int \left(\frac{u^2+1}{1+u^2} - \frac{1}{1+u^2} \right) du$$

$$= 2 \int \left(1 - \frac{1}{1+u^2} \right) du$$

$$= 2u - 2 \tan^{-1}(u) + c$$

$$= 2\sqrt{x} - 2 \tan^{-1}(\sqrt{x}) + c \quad \checkmark$$

(b)

* Rewrite problem as:

Show that $4^{2n+1} + 5^{2n+1} + 6^{2n+1}$ is a multiple of 15 for all $n \geq 0$.

Prove true for $n=0$:

$$4^1 + 5^1 + 6^1 = 15$$

$\therefore$ true when $n=0$. $\checkmark$

Assume true for some $n=k$, where $k \geq 0$.

i.e. $4^{2k+1} + 5^{2k+1} + 6^{2k+1} = 15M$, where $M \in \mathbb{Z}$.

Prove true for $n=k+1$:

RTP: $4^{2k+3} + 5^{2k+3} + 6^{2k+3}$ is a multiple of 15.

$$\begin{aligned} 4^{2k+3} + 5^{2k+3} + 6^{2k+3} &= 4^2 \times 4^{2k+1} + 5^2 \times 5^{2k+1} + 6^2 \times 6^{2k+1} \\ &= 16(15M - 5^{2k+1} - 6^{2k+1}) + 25 \times 5^{2k+1} + 36 \times 6^{2k+1} \\ &\quad \text{(from assumption)} \quad \checkmark \\ &= 16 \times 15M + 9 \times 5^{2k+1} + 20 \times 6^{2k+1} \\ &= 16 \times 15M + 45 \times 5^{2k} + 120 \times 6^{2k} \\ &= 15(16M + 3 \times 5^{2k} + 8 \times 6^{2k}) \quad \checkmark \\ &= 15N, \text{ where } N \in \mathbb{Z} \text{ since } k \geq 0 \end{aligned}$$

$\therefore 4^n + 5^n + 6^n$ is a multiple of 15 for all odd $n \geq 1$.

(15)(c)(i)

$$\ddot{y} = 0 \text{ when travelling at terminal velocity.}$$
$$\therefore 10m - mkv_T = 0$$
$$v_T = \frac{10}{k} \text{ ms}^{-1} \quad \checkmark$$

(ii)(a)

$$m \ddot{y} = 10m - mkv$$

$$\ddot{y} = 10 - kv$$

$$v \cdot \frac{dv}{dy} = 10 - kv$$

$$\frac{dy}{dv} = \frac{v}{10 - kv}$$

$$= -\frac{1}{k} \times \frac{-kv + 10 - 10}{10 - kv}$$

$$\int -k dy = \int \left(1 - \frac{10}{10 - kv} \right) dv \quad \checkmark$$

$$-ky = \left(v + \frac{10}{k} \ln |10 - kv| \right) + C \quad \checkmark$$

(correctly rearranging
as separable DE
or equivalent)

When $y=0$, $v = \frac{20}{k}$

$$\therefore C = -\frac{20}{k} - \frac{10}{k} \ln \left| 10 - k \times \frac{20}{k} \right|$$
$$= -\frac{20}{k} - \frac{10}{k} \ln 10$$

$$\therefore -ky = v + \frac{10}{k} \ln |10 - kv| - \frac{20}{k} - \frac{10}{k} \ln 10$$

$$k^2 y = -kv - 10 \ln |10 - kv| + 20 + 10 \ln 10$$

$$= 20 - kv + 10 \ln \left| \frac{10}{10 - kv} \right|$$

$$\therefore y = \frac{1}{k^2} \left(20 - kv + 10 \ln \left| \frac{10}{10 - kv} \right| \right) \quad \checkmark$$

(b)

When $y=50$, $v = \frac{3}{4} \times \frac{20}{k}$

$$= \frac{15}{k} \quad \checkmark$$

$$50 = \frac{1}{k^2} \left(20 - k \times \frac{15}{k} + 10 \ln \left| \frac{10}{10 - k \times \frac{15}{k}} \right| \right)$$

$$50 = \frac{1}{k^2} (20 - 15 + 10 \ln 2)$$

$$k^2 = \frac{5 + 10 \ln 2}{50}, \quad \therefore k = 0.49 \quad \checkmark \quad (2 \text{ d.p.})$$

15(d)(i)

$$r_1 = \begin{bmatrix} -\lambda \\ 5+4\lambda \\ 4+3\lambda \end{bmatrix}, \therefore \vec{OA} = \begin{bmatrix} -p \\ 5+4p \\ 4+3p \end{bmatrix}$$

$$r_2 = \begin{bmatrix} -2-\mu \\ 4+2\mu \\ 1+2\mu \end{bmatrix}, \therefore \vec{OB} = \begin{bmatrix} -2-q \\ 4+2q \\ 1+2q \end{bmatrix}$$

$$\vec{AB} = \vec{OB} - \vec{OA}$$

$$= \begin{bmatrix} -2-q+p \\ -1+2q-4p \\ -3+2q-3p \end{bmatrix} \quad \checkmark$$

(ii) $|\vec{AB}|$ is a minimum when $\vec{AB}$ is perpendicular to both lines.

$$\vec{AB} \cdot \begin{bmatrix} -1 \\ 4 \\ 3 \end{bmatrix} = 0: \quad \begin{aligned} 2+q-p-4+8q-16p-9+6q-9p &= 0 \\ -11+15q-26p &= 0 \end{aligned} \quad (1)$$

$$\vec{AB} \cdot \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix} = 0: \quad \begin{aligned} 2+q-p-2+4q-8p-6+4q-6p &= 0 \\ -6+9q-15p &= 0 \end{aligned}$$

$$3q = 5p + 2 \quad (2)$$

$$\text{sub. into (1):} \quad \begin{aligned} -11+5(5p+2)-26p &= 0 \\ -11+25p+10-26p &= 0 \end{aligned}$$

$$\text{sub. into (2):} \quad \begin{aligned} p &= -1 \\ 3q &= -5+2 \\ &= -3 \end{aligned}$$

$$\begin{aligned} \text{When } p=-1, q=-1: \\ \vec{AB} &= \begin{bmatrix} -2-(-1)-1 \\ -1+2(-1)-4(-1) \\ -3+2(-1)-3(-1) \end{bmatrix} \\ &= \begin{bmatrix} -2 \\ 1 \\ -2 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \text{and } |\vec{AB}| &= \sqrt{(-2)^2 + 1^2 + (-2)^2} \\ &= \sqrt{9} \\ &= 3 \text{ units} \end{aligned} \quad \checkmark$$

(15) (d) (iii) Note that $|\vec{AB}|$ is a minimum when $\vec{AB}$ is perpendicular to both $\vec{r}_1$ and $\vec{r}_2$.

As such, $|\vec{AB}| \geq 3$ ✓ (with equality when $p=-1$ and $q=-1$)

(16)(a)(i)

$$\int_0^{\pi} x f(\sin x) dx$$

Let $x = \pi - u$

$dx = -du$

When $x=0$, $u=\pi$

$x=\pi$, $u=0$

$$= \int_{\pi}^0 (\pi - u) f(\sin(\pi - u)) (-1) du$$

$$= \int_0^{\pi} (\pi - u) f(\sin u) du \quad \checkmark \quad \text{since } \sin(\pi - u) = \sin u$$

$$= \pi \int_0^{\pi} f(\sin u) du - \int_0^{\pi} u f(\sin u) du$$

$$= \pi \int_0^{\pi} f(\sin x) dx - \int_0^{\pi} x f(\sin x) dx \quad (\text{dummy variables})$$

$$\therefore 2 \int_0^{\pi} x f(\sin x) dx = \pi \int_0^{\pi} f(\sin x) dx$$

$$\int_0^{\pi} x f(\sin x) dx = \frac{\pi}{2} \int_0^{\pi} f(\sin x) dx \quad \checkmark$$

(ii) $\int_0^{\pi} (1+2x) \frac{\sin^3 x}{1+\cos^2 x} dx = \int_0^{\pi} \frac{\sin^3 x dx}{1+\cos^2 x} + 2 \int_0^{\pi} x \frac{\sin^3 x dx}{1+\cos^2 x}$

as $\frac{\sin^3 x}{1+\cos^2 x} = \frac{\sin^3 x}{1+(1-\sin^2 x)}$, using result from part (i):

$$= \int_0^{\pi} \frac{\sin^3 x dx}{1+\cos^2 x} + 2 \times \frac{\pi}{2} \int_0^{\pi} \frac{\sin^3 x dx}{1+\cos^2 x} \quad \checkmark$$

$$= (1+\pi) \int_0^{\pi} \frac{\sin x \cdot (1-\cos^2 x) dx}{1+\cos^2 x}$$

$$= (1+\pi) \int_1^{-1} \frac{1-u^2}{1+u^2} (-du) \quad \checkmark \quad \begin{array}{l} \text{Let } u = \cos x \\ du = -\sin x dx \\ x=0, u=1 \\ x=\pi, u=-1 \end{array}$$

$$= (1+\pi) \int_{-1}^1 \frac{1-u^2}{1+u^2} du$$

$$= (1+\pi) \int_{-1}^1 \left(\frac{-(1+u^2) + 2}{1+u^2} \right) du$$

$$= (1+\pi) \int_{-1}^1 \left(-1 + \frac{2}{1+u^2} \right) du$$

$$\begin{aligned}
 &= (1+\pi) \left[-u + 2 \tan^{-1}(u) \right]_{-1}^1 \\
 &= (1+\pi) \left(-1 + \frac{\pi}{2} - \left(1 - \frac{\pi}{2} \right) \right) \\
 &= (1+\pi)(\pi-2) \quad \checkmark
 \end{aligned}$$

16(b)(i)

$$\begin{aligned}
 x &= A e^{-kt} \cos nt \\
 \dot{x} &= -k A e^{-kt} \cos nt - n A e^{-kt} \sin nt \\
 \ddot{x} &= k^2 A e^{-kt} \cos nt + nk A e^{-kt} \sin nt + nk A e^{-kt} \sin nt - n^2 A e^{-kt} \cos nt \\
 &= (A k^2 - A n^2) e^{-kt} \cos nt + (2 A n k) e^{-kt} \sin nt \quad \checkmark \text{ (1)}
 \end{aligned}$$

$$\begin{aligned}
 \text{Also, } \ddot{x} &= -P^2 x - Q \dot{x} \\
 &= -P^2 A e^{-kt} \cos nt - Q (-k A e^{-kt} \cos nt - n A e^{-kt} \sin nt) \\
 &= (-A P^2 + A k Q) e^{-kt} \cos nt + A n Q e^{-kt} \sin nt \quad \checkmark \text{ (2)}
 \end{aligned}$$

As the expressions in the RHS of (1) and (2) must be equivalent, equate coefficients of $e^{-kt} \cos nt$ and $e^{-kt} \sin nt$:

$$2 A n k = A n Q$$

$$\therefore k = \frac{1}{2} Q$$

$$A k^2 - A n^2 = -A P^2 + A k Q$$

$$\left(\frac{1}{2} Q \right)^2 - n^2 = -P^2 + \frac{1}{2} Q^2$$

$$n^2 = P^2 - \frac{1}{4} Q^2$$

$$= \frac{1}{4} (4 P^2 - Q^2)$$

$$\therefore n = \frac{1}{2} \sqrt{4 P^2 - Q^2} \quad \checkmark$$

Also note that when $t=0$, $x = A e^0 \times \cos 0$
 $= A$

So the initial displacement is also satisfied.

(16) contd.

(b)(ii) Particle is at rest when $\dot{x} = 0$:

$$-Ake^{-kt} \cos nt - Ane^{-kt} \sin nt = 0$$

$$n \sin nt = -k \cos nt$$

$$\tan nt = -\frac{k}{n}$$

As $k, n > 0$, $-\frac{k}{n} < 0$. $\therefore$ for positive t :

$$nt = \pi - \tan^{-1}\left(\frac{k}{n}\right), 2\pi - \tan^{-1}\left(\frac{k}{n}\right), 3\pi - \tan^{-1}\left(\frac{k}{n}\right), \dots$$

$$t = \frac{1}{n}(\pi - \alpha), \frac{1}{n}(2\pi - \alpha), \frac{1}{n}(3\pi - \alpha), \dots, \text{ where } \alpha = \tan^{-1}\left(\frac{k}{n}\right)$$

Particle first comes to rest when $t = \frac{1}{n}(\pi - \alpha)$

$$\therefore x_1 = Ae^{-k \times \frac{1}{n}(\pi - \alpha)} \cos\left(n \times \frac{1}{n}(\pi - \alpha)\right)$$

$$= Ae^{-\frac{k\pi}{n} + \frac{k\alpha}{n}} \cos(\pi - \alpha)$$

$$= -Ae^{\frac{k\alpha}{n}} \cos \alpha \times e^{-\frac{k\pi}{n}}, \text{ since } \cos(\pi - \alpha) = -\cos \alpha$$

(iii) When $t = \frac{1}{n}(2\pi - \alpha)$:

$$x_2 = Ae^{-k \times \frac{1}{n}(2\pi - \alpha)} \cos\left(n \times \frac{1}{n}(2\pi - \alpha)\right)$$

$$= Ae^{\frac{k\alpha}{n}} \cos(2\pi - \alpha) \times e^{-\frac{2k\pi}{n}}$$

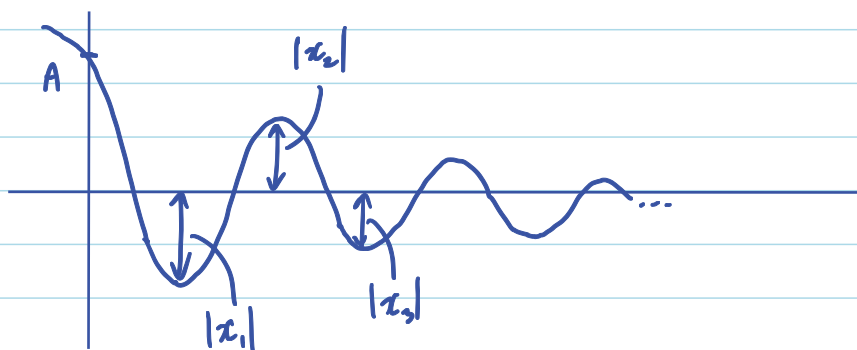
$$= Ae^{\frac{k\alpha}{n}} \cos \alpha \times e^{-\frac{2k\pi}{n}} \text{ since } \cos(2\pi - \alpha) = \cos \alpha$$

(some correct exploration of x_2)

Note that for successive values of t , $\cos nt$ will alternate between $\cos \alpha$ and $-\cos \alpha$. Each successive value of x_r can be found by multiplying the previous by $-e^{-\frac{k\pi}{n}}$.

$$\therefore |x_r| = |x_{r-1}| \times e^{-\frac{k\pi}{n}}, \text{ forming a G.P.}$$

(recognising geometric progression)



Let total distance travelled by D metres.

$$\begin{aligned} \text{Then } D &= A + 2|x_1| + 2|x_2| + 2|x_3| + \dots \\ &= A + 2(|x_1| + |x_2| + |x_3| + \dots) \end{aligned}$$

$$= A + 2 \times \frac{A e^{\frac{kx}{n}} \cos \alpha e^{-\frac{k\pi}{n}}}{1 - e^{-\frac{k\pi}{n}}}$$



$$= A \left(1 + \frac{2A e^{\frac{kx}{n}} \cos \alpha}{e^{\frac{k\pi}{n}} - 1} \right)$$

$$\text{where } \alpha = \tan^{-1}\left(\frac{k}{n}\right)$$

Note that for a given A , D is dependent on $\frac{k}{n}$.

$$\text{Now } \frac{k}{n} = \frac{\frac{1}{2}Q}{\frac{1}{2}\sqrt{4P^2 - Q^2}}$$

$$= \frac{1}{\sqrt{\frac{4P^2 - Q^2}{Q^2}}}$$

$$= \frac{1}{\sqrt{4\left(\frac{P}{Q}\right)^2 - 1}}$$

$\therefore$ For a given A , D can be expressed as a function of $\frac{k}{n}$, and $\frac{k}{n}$ can be expressed as a function of $\frac{P}{Q}$.
 $\therefore$ The total distance travelled depends only on $\frac{P}{Q}$. ✓